Task 2 under Bernoulli Labels and Logit-Space Annotation Bias

Complete corrected framework and derivation

Scope. This note gives the Bernoulli version of Task 2 when the available references are trained probability models. The proxy distortion is controlled directly. The only Gaussian assumption is imposed on the global logit offsets $C_r = \mathbb{E}_X b_r(X)$; no Gaussian assumption is imposed on g_r or $g_{\widehat{R}}$.

1 Bernoulli/logit model

For reference annotator r,

$$Y_r(x) \mid x, \xi_r \sim \text{Bernoulli}\{\mu_r(x)\}, \qquad \text{logit}\{\mu_r(x)\} = \text{logit}\{\mu^*(x)\} + b_r(x). \tag{1}$$

Here $\sigma(t) = 1/(1+e^{-t})$, $\mathbb{E}_r$ averages over a random reference identity, and $\mathbb{E}_X$ averages over the item distribution. Also $\mathbb{E}[Y_B \mid X] = \mu_B(X)$. The reference identities are i.i.d. draws from the annotator population and $R \cap \{A, B\} = \varnothing$. Conditional on X and the annotator identities, labels from distinct annotators are independent. For every fixed x,

$$\mathbb{E}_r b_r(x) = 0, \qquad \mathbb{E}_r b_r(x)^2 < \infty.$$

The global signed logit offset satisfies

$$C_r := \mathbb{E}_X b_r(X), \qquad C_r \overset{\text{i.i.d.}}{\sim} N(0, \tau^2).$$

This is the only normal random-effect model used below.

The sensitivity condition is

$$\mathbb{E}_r \operatorname{Var}_X\{b_r(X)\} \le (\kappa - 1)\mathbb{E}_r\{\mathbb{E}_X b_r(X)\}^2 = (\kappa - 1)\tau^2, \qquad \kappa \ge 1. \tag{2}$$

Consequently,

$$\mathbb{E}_X \mathbb{E}_r b_r(X)^2 = \mathbb{E}_r \operatorname{Var}_X\{b_r(X)\} + \mathbb{E}_r\{\mathbb{E}_X b_r(X)\}^2 \le \kappa\tau^2.$$

The predictors $\widehat{\mu}_A$ and $\widehat{\mu}_r$ are trained by Bernoulli cross-entropy and output probabilities. Population cross-entropy is minimized by the corresponding conditional probability, so $\widehat{\mu}_r$ is intended to estimate μ_r, not the latent μ^*.

2 Trained-reference replacement

Because b_r is now a logit-space bias, the correct decomposition is

$$\widehat{\mu}_r(x) = \mu_r(x) + h_r(x) = \sigma(\text{logit}\{\mu^*(x)\} + b_r(x)) + h_r(x),$$

not $\widehat{\mu}_r = \mu^* + b_r + h_r$. Define, as in the original trained-reference construction,

$$\widehat{\mu}_R(x) = \frac{1}{K}\sum_{r \in R} \widehat{\mu}_r(x), \qquad \overline{h}_R(x) = \frac{1}{K}\sum_{r \in R} h_r(x).$$

Assumption 1 (Centered trained-reference error)**.** $\mathbb{E}[\overline{h}_R(X)] = 0$.

Assumption 2 (Orthogonality of trained-reference error)**.** $\overline{h}_R(X) \perp\!\!\!\perp \{\widehat{\mu}_A(X) - Y_B(X)\}$.

These two assumptions imply

$$\mathbb{E}\big[\overline{h}_R(X)\{\widehat{\mu}_A(X) - Y_B(X)\}\big] = 0.$$

They are imposed on the probability-scale error $h_r = \widehat{\mu}_r - \mu_r$. Reference-model training data must be disjoint from D_1 and D_2, or the predictions on D_1, D_2 must be cross-fitted. The sets D_1 and D_2 themselves need not be independent.

Define

$$\Delta^* := \mathbb{E}\big[\{\widehat{\mu}_A(X) - \mu^*(X)\}^2 - \{Y_B(X) - \mu^*(X)\}^2\big],$$
$$\Delta_{\widehat{R}} := \mathbb{E}\big[\{\widehat{\mu}_A(X) - \widehat{\mu}_R(X)\}^2 - \{Y_B(X) - \widehat{\mu}_R(X)\}^2\big],$$

and

$$g_{\widehat{R}} := \mathbb{E}[\{\widehat{\mu}_R(X) - \mu^*(X)\}\{\widehat{\mu}_A(X) - Y_B(X)\}].$$

Expanding the two squared-loss differences gives the exact identity

$$\Delta_{\widehat{R}} - \Delta^* = -2g_{\widehat{R}}, \qquad \operatorname{sign}(\Delta_{\widehat{R}}) = \operatorname{sign}(\Delta^*) \neq 0 \iff \Delta_{\widehat{R}}(\Delta_{\widehat{R}} + 2g_{\widehat{R}}) > 0. \tag{3}$$

For each reference annotator, retain the original notation

$$g_r := \mathbb{E}_X[\{\mu_r(X) - \mu^*(X)\}\{\widehat{\mu}_A(X) - \mu_B(X)\}].$$

Conditional independence removes the Bernoulli label noise but does not remove the common item X. Hence

$$\mathbb{E}[(Y_r - \mu^*)(\widehat{\mu}_A - Y_B)] = g_r,$$

and the expectation over X cannot be split into a product of two expectations. The trained-reference assumptions give

$$g_{\widehat{R}} = \frac{1}{K}\sum_{r\in R} g_r + \mathbb{E}\big[\overline{h}_R(X)\{\widehat{\mu}_A(X) - Y_B(X)\}\big] = \frac{1}{K}\sum_{r\in R} g_r.$$

Thus it is enough to control the complete quantity $K^{-1}\sum_{r\in R} g_r$ directly.

3 Logit-space scaling and the Bernstein bound

Taylor's theorem gives

$$\mu_r(X) - \mu^*(X) = \sigma'(\operatorname{logit}\mu^*(X))\, b_r(X) + \frac{1}{2}\sigma''(\cdot) b_r(X)^2.$$

The first-order term vanishes after averaging over the reference identity because $\mathbb{E}_r b_r(X) = 0$. Since

$$\sup_t |\sigma''(t)| = \frac{\sqrt{3}}{18}, \qquad |\widehat{\mu}_A(X) - \mu_B(X)| \le 1,$$

the annotator-population mean obeys

$$|\mathbb{E}_r g_r| \le \frac{1}{2}\sup_t |\sigma''(t)|\, \mathbb{E}_X \mathbb{E}_r b_r(X)^2 \le \frac{\sqrt{3}}{36}\kappa\tau^2.$$

The coefficient $\sqrt{3}/36$ is the sharp uniform quadratic coefficient obtainable from the sigmoid without restricting the relevant logit range.

Next, the sigmoid is 1/4-Lipschitz. Jensen's inequality therefore yields

$$\operatorname{Var}_r(g_r) \le \mathbb{E}_r g_r^2 \le \mathbb{E}_X \mathbb{E}_r\big[(\mu_r - \mu^*)^2(\widehat{\mu}_A - \mu_B)^2\big] \le \frac{\kappa\tau^2}{16}.$$

Moreover, for any two reference identities r, s,

$$|g_r - g_s| \le \mathbb{E}_X[|\mu_r - \mu_s|\,|\widehat{\mu}_A - \mu_B|] \le 1.$$

Hence $|g_r - \mathbb{E}_r g_r| \le 1$ and also $\operatorname{Var}_r(g_r) \le 1/4$. Bernstein's inequality then implies that, on any event $\tau \le U_{\tau,\gamma}(|D_2|)$, with probability at least $1 - \alpha_g$ over the K reference identities,

$$|g_{\widehat{R}}| \le \min\left\{1, \frac{\sqrt{3}}{36}\kappa U_{\tau,\gamma}(|D_2|)^2 + \sqrt{\frac{\min\{\kappa U_{\tau,\gamma}(|D_2|)^2, 4\}\log(2/\alpha_g)}{8K}} + \frac{\log(2/\alpha_g)}{3K}\right\}. \tag{4}$$

The additional minimum inside the square root uses the range-based variance bound 1/4. The term $1/(3K)$ is the standard sub-gamma Bernstein constant. No distributional assumption on g_r is used.

4 Calibration of τ from trained reference probabilities

Assume $K > 1$. The logit model implies

$$C_r - C_1 = \mathbb{E}_X\{b_r(X) - b_1(X)\} = \mathbb{E}_X[\operatorname{logit}\mu_r(X) - \operatorname{logit}\mu_1(X)]\,.$$

Therefore μ^* is not needed: it cancels in pairwise logit differences. For $r = 2, \ldots, K$, compute on D_2

$$\overline{w}_r := \frac{1}{|D_2|}\sum_{i\in D_2}\{\operatorname{logit}\widehat{\mu}_r(x_i) - \operatorname{logit}\widehat{\mu}_1(x_i)\}\,, \qquad \overline{w} = (\overline{w}_2, \ldots, \overline{w}_K)^\top.$$

Let $\widehat{\Omega}$ be the sample covariance matrix on D_2 of the vector

$$\big(\operatorname{logit}\widehat{\mu}_2(X) - \operatorname{logit}\widehat{\mu}_1(X), \ldots, \operatorname{logit}\widehat{\mu}_K(X) - \operatorname{logit}\widehat{\mu}_1(X)\big)^\top.$$

Because $C_r \overset{\text{i.i.d.}}{\sim} N(0, \tau^2)$,

$$\operatorname{Var}(C_r - C_1) = 2\tau^2, \qquad \operatorname{Cov}(C_r - C_1, C_s - C_1) = \tau^2 \quad (r \neq s).$$

For a candidate $t \geq 0$, define $\widehat{V}_\tau(t)$ by

$$\widehat{V}_\tau(t)_{rr} = 2t^2 + \frac{\widehat{\Omega}_{rr}}{|D_2|}, \qquad \widehat{V}_\tau(t)_{rs} = t^2 + \frac{\widehat{\Omega}_{rs}}{|D_2|} \quad (r \neq s),$$

where $r, s = 2, \ldots, K$, and set

$$\widehat{Q}_\tau(t) := \overline{w}^\top \widehat{V}_\tau(t)^{-1}\overline{w}.$$

Let $q_{\gamma,K-1}$ be the lower γ-quantile of χ^2_{K-1}. The upper endpoint is

$$U_{\tau,\gamma}(|D_2|) := \sup\{t \geq 0 : \widehat{Q}_\tau(t) \geq q_{\gamma,K-1}\}, \tag{5}$$

with the value zero if the set is empty. Since $\widehat{Q}_\tau(t)$ is non-increasing in t, the fixed-K calibration gives

$$\mathbb{P}\{\tau \leq U_{\tau,\gamma}(|D_2|)\} = 1 - \gamma + o(1).$$

Indeed, under the conditions below,

$$\widehat{Q}_\tau(\tau) \Rightarrow \chi^2_{K-1},$$

and

$$\{\tau \leq U_{\tau,\gamma}(|D_2|)\} = \{\widehat{Q}_\tau(\tau) \geq q_{\gamma,K-1}\}.$$

Conditions required of $\widehat{\mu}_r$ for this calibration

The two trained-reference assumptions in Section 2 justify replacing Y_R by $\widehat{\mu}_R$ in the probability-scale distortion. They do not by themselves justify applying logit to $\widehat{\mu}_r$. For (5), require:

1. $0 < \widehat{\mu}_r(x) < 1$ on D_2. If numerical clipping is used, its contribution must satisfy the logit-contrast accuracy condition below.
2. Each $\widehat{\mu}_r$ is evaluated out of sample on D_2 and estimates the Bernoulli conditional probability $\mu_r(x) = \mathbb{P}(Y_r = 1 \mid X = x, \xi_r)$.
3. For every $r = 2, \ldots, K$,

$$\mathbb{E}_X[\operatorname{logit}\widehat{\mu}_r(X) - \operatorname{logit}\widehat{\mu}_1(X)] = C_r - C_1 + o_p(|D_2|^{-1/2}). \tag{6}$$

   This is the required first-stage bias condition. A sufficient, stronger condition is logit-scale consistency of each $\widehat{\mu}_r$ for μ_r at the same rate. Probability-scale consistency alone is insufficient near zero or one because the derivative of $\operatorname{logit}(p)$ is $1/\{p(1-p)\}$.
4. The fixed-dimensional multivariate CLT holds for $\overline{w}$, $\widehat{\Omega}$ consistently estimates its item-sampling covariance, and $\widehat{V}_\tau(t)$ is nonsingular where it is inverted. If model-training uncertainty is not negligible at the $|D_2|^{-1/2}$ scale, its covariance must be added; otherwise (5) is not a valid upper endpoint.

The order of operations is essential: first take each signed average $\overline{w}_r$ over D_2, and then calibrate their between-annotator scale. The absolute values, RMS, or quantiles of the pointwise logit differences estimate itemwise variation and must not be used as $U_{\tau,\gamma}$.

5 Main statistic, test, and validity

On D_1, compute

$$\widehat{\Delta}_{\widehat{R}} := \frac{1}{|D_1|} \sum_{j \in D_1} \left[\{\widehat{\mu}_A(x_j) - \widehat{\mu}_R(x_j)\}^2 - \{y_{B,j} - \widehat{\mu}_R(x_j)\}^2 \right]$$

and its paired standard error $\widehat{\mathrm{se}}(\widehat{\Delta}_{\widehat{R}})$. Test

$$H_0 : \Delta_{\widehat{R}}(\Delta_{\widehat{R}} + 2g_{\widehat{R}}) \le 0, \qquad H_1 : \Delta_{\widehat{R}}(\Delta_{\widehat{R}} + 2g_{\widehat{R}}) > 0.$$

Reject H_0 when

$$\begin{aligned} &|\widehat{\Delta}_{\widehat{R}}| - t_{|D_1|-1,1-\alpha_\Delta/2}\, \widehat{\mathrm{se}}(\widehat{\Delta}_{\widehat{R}}) \\ &\quad > 2 \min\left\{ 1, \frac{\sqrt{3}}{36} \kappa U_{\tau,\gamma}(|D_2|)^2 + \sqrt{\frac{\min\{\kappa U_{\tau,\gamma}(|D_2|)^2, 4\} \log(2/\alpha_g)}{8K}} + \frac{\log(2/\alpha_g)}{3K} \right\}. \end{aligned} \tag{7}$$

Theorem 1 (False-certification control)**.** *Assume (1)–(2), the trained-reference conditions in Section 2, the calibration conditions in Section 4, and the paired CLT and standard-error consistency for $\widehat{\Delta}_{\widehat{R}}$. For fixed K, under H_0,*

$$\limsup_{|D_1|,|D_2| \to \infty} \mathbb{P}(\textit{reject } H_0) \le \alpha_\Delta + \alpha_g + \gamma.$$

Proof. The paired CLT gives, with probability $1 - \alpha_\Delta + o(1)$,

$$|\widehat{\Delta}_{\widehat{R}}| - t_{|D_1|-1,1-\alpha_\Delta/2} \widehat{\mathrm{se}}(\widehat{\Delta}_{\widehat{R}}) \le |\Delta_{\widehat{R}}|.$$

Under H_0, zero lies between $\Delta_{\widehat{R}}$ and $\Delta_{\widehat{R}} + 2g_{\widehat{R}}$, so

$$|\Delta_{\widehat{R}}| \le 2|g_{\widehat{R}}|.$$

The calibration event in (5) has probability $1 - \gamma + o(1)$. On that event, (4) holds with probability at least $1 - \alpha_g$. Therefore rejection under H_0 requires failure of at least one of these three events. A union bound proves the result. No independence among the three events is required. □

6 Implementation order

1. Train $\widehat{\mu}_A$ and each $\widehat{\mu}_r$ on their annotator-specific Bernoulli labels using BCE; evaluate them out of sample.
2. On D_2, compute the signed pairwise logit averages $\overline{w}_r$, the full sample covariance $\widehat{\Omega}$, and $U_{\tau,\gamma}(|D_2|)$ from (5).
3. Specify κ through (2).
4. On D_1, compute $\widehat{\mu}_R$, $\widehat{\Delta}_{\widehat{R}}$, and its paired standard error.
5. Apply (7).

The curvature term in (4) does not decrease with K; the two Bernstein panel terms decrease at rates $K^{-1/2}$ and K^{-1}. This is the direct consequence of passing centered logit bias through the nonlinear sigmoid.